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Abstract

This is also undecided at this time and will be filled in later.

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Rethink the structure of the background chapter. Possibly reorder it	5
Fix up the figure to make the multi headed attention proper and clearer.	6
There is something here yet I am not quite sure. One could change the axes in 2.2 to represent the concept of actual values and perceived values (however this feels like trying to shoehorn it in too much when the concept of revealed preferences is more appropriate). Another idea is that prospect theory explains how users will provide feedback in a biased way and we should actually be using the prospect theory to map that to the true thing we should align for, might help with generalisation.	12
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Chapter 1

Introduction

Unsure if this is needed. Yet could be a simple "one pager" (which is bigger than the one paragraph of the abstract) that introduces the topic area, motivates the project and outlines the project.

AI systems are on a trajectory to become more ingrained and operate more autonomously in our society. As these systems —currently LLMs— will increasingly make more decisions that effect human lives it is crucial that these systems are aligned with and reflect the values of the people they serve. What is being built could be your plumber, your lawyer or your doctor tomorrow what kind of being do you want these systems to be?

SOME paragraph introducing the concept of alignment. Then the concept of AI alignment

Chapter 2

Background

Rethink the structure of the background chapter. Possibly reorder it

2.1 Large Language Models

2.1.1 Language models

Language models (LMs) are a class of machine learning models that are designed to understand and generate human language. They are typically probabilistic models that learn to predict the next token in a sequence of text given the preceding context. By training on large datasets of text data (corpora), language models can learn the statistical patterns and structures inherent in human language, enabling them to generate coherent and contextually relevant text.

Language models see the world as tokens which are generated by taking a string and chunking it into smaller pieces. Tokens can represent words, parts of words, or even individual characters depending on the tokenization method used. Most language models today use Byte pair encoding [79] that breaks text into subwords. Almost all successful language models use probabilistic methods, therefore they are models of the form $P(t_i | t_1, t_2, \dots, t_{i-1})$ where t_i is the token to be predicted and $t_1, t_2, \dots, t_{i-1}$ are the preceding tokens in the sequence. Earlier methods of rule based systems failed due to the complexity and ambiguity of human language and rules being too rigid [72, 65].

Early probabilistic models included simple statistical approaches such as n-grams, which relied on counting the frequency of token sequences. However this approach suffered from two competing issues, short context windows (2-grams, 3-grams etc) means that model can't understand long range dependencies in text, while larger n-grams leads to data sparsity issues where many long sequences are never seen in training data. The advent of neural networks allowed for more sophisticated models such as Feedforward Neural Language models [14], Recurrent Neural Networks (RNNs) [100] and Long Short-Term Memory networks (LSTMs) [68] which could in theory capture longer range dependencies in text. However these models suffered from issues such as

- **Vanishing gradients:** Early tokens in a long sequence have little effect on gradients and therefore the models understanding of long range dependencies.
- **Information bottlenecks:** The hidden state of the model has to compress all prior context into a fixed size vector.

Crucially these models failed in their flexibility to model the complex dependencies in human language.

2.1.2 Modern attention based Large Language Models

The key feature in language is context, and it is this context of surrounding words that let us humans understand and break through the complexity and ambiguity of language. The challenge is that the way in which a word's company affects the meaning of the word is complex, therefore any language model needs to be able effectively capture these complex relationships. Attention is the concept where each token's representation is influenced by all the other tokens in the sequence (or just the preceding tokens in the case of autoregressive models). This was first applied to RNN-based LM in [10], however in [104] it was shown that attention mechanisms alone were sufficient to model language without the need for any recurrent structure¹.

Transformers [104] are a class of neural network architecture that utilizes the attention mechanism to process data and generate a concise representation (encoder-only), or generate data autoregressively (decoder-only). The canonical transformer architecture consists of two parts an encoder and a decoder, however decoder only architectures are shown to be sufficient for all current generative language modelling tasks. A simplified architecture of a decoder-only transformer model is shown in Figure 2.1. The key feature of the transformer is the self attention. Each token has an embedding vector (used to represent its context aware meaning, i.e. river *bank* vs *bank* robbery) which is iteratively updated by each layer of self-attention. The self-attention mechanism involves matrix multiplication between the current token and other tokens in the sequence. For generative tasks an attention mask is applied so that each token can only 'see' preceding tokens in the sequence. Furthermore to allow for different types of dependencies to be captured this attention is done multi times in parallel (multi-head attention) and merged together at the end of each layer. After several layers of self-attention the final set of token embeddings are used to generate a probability distribution over the vocabulary for the next token. A token can be sampled from this distribution, appended to the input sequence and the process repeated to generate long sequences of text.

Modern transformer based large language models (LLMs) such as GPT-3 [21] are some of the largest machine learning models ever created with hundreds of billions of parameters and now trillions of parameters. Therefore training these models requires massive datasets. They are trained in a self supervised manner on large corpora (hundreds of terabytes) of text data from the internet such as Common Crawl [28], WebText [38], and others. The objective function is usually the cross-entropy loss between the predicted token distribution and the actual next token in the sequence. The output of this training process is a model that can generate coherent and contextually relevant text given a prompt.

2.1.3 LLM in the modern world

Large language models (LLMs) are fundamentally just language models which can predict the next token in a sequence of text. However due to the prevalence of natural language in our world and the capabilities of LLMs they have applications in a wide variety of domains. This includes Virtual Assistants, Content creation, Code generation, Translation and many more. Furthermore the transformer architecture has been demonstrated to be effective in other domains like computer vision [64] which has lead to multi-modal models that can process and generate text, images, and other data types [3]. This flexibility of LLMs have lead to them have widespread adoption and proliferation in modern information systems

¹Recurrent structure is where there is a hidden state (normally a fixed length vector) that is passed along and updated each time by the next input. For language modelling this hidden state is supposed to contain the context needed to understand the next input token. However as discussed above it was not sufficient.

Fix up the figure to make the multi headed attention proper and clearer.

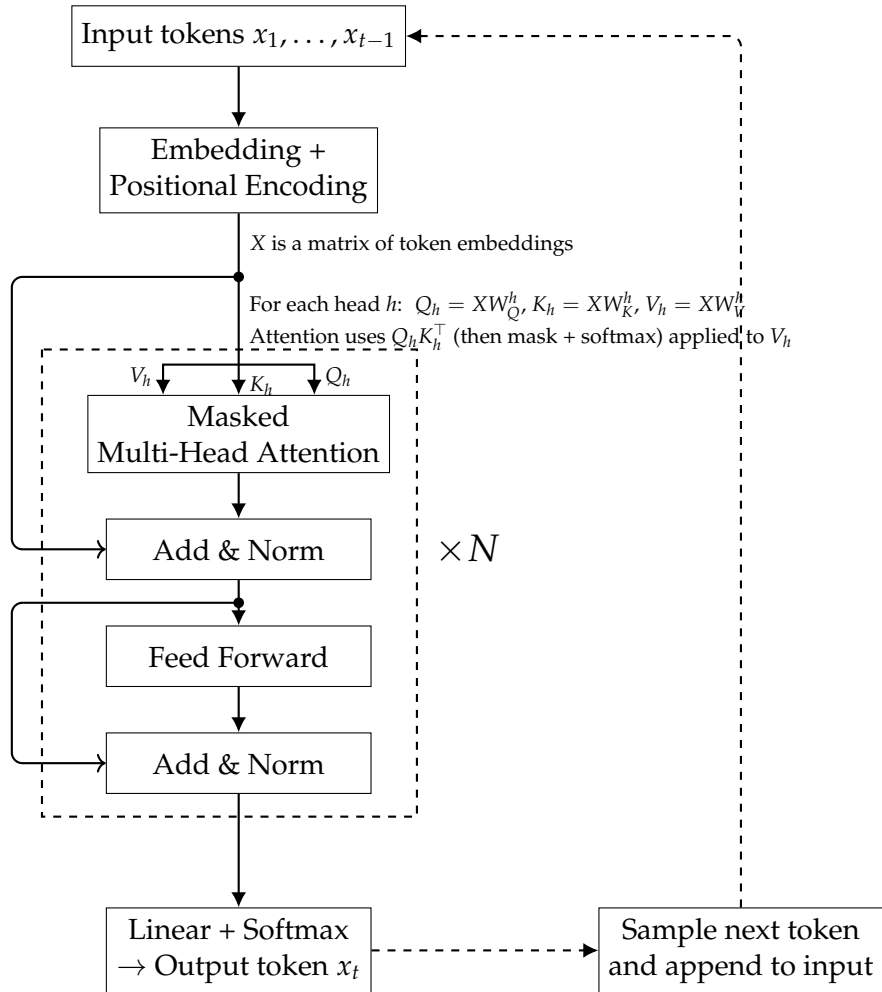


Figure 2.1: Simplified architecture of a decoder-only transformer model for language modeling. Adapted from [104].

². Along with this proliferation there has been increasing concern about the safety and ethical implications of these models. Specifically these models have been shown to exhibit concerning behaviors like self-preservation and deception [67, 40, 25]. On top of this there are growing ethical questions around its use in our society [13, 15] as well as concerns of the welfare of the models themselves [23].

2.2 Alignment of Large Language Models

2.2.1 What is alignment?

As discussed above large language models (LLMs) have shown remarkable capabilities in understanding and generating human-like text. However the pre-training process focuses primarily on predicting the next token in a sequence, this does not necessarily mean that the model will produce useful, safe outputs that align with what users or designers want. Therefore, aligning LLMs is generally ³ the process of taking the pre-trained model and fine-tuning it to better match what the designers and end users expect from the model. The aligning process can work on many dimensions of the models behavior from helpfulness, politeness, honesty, and safety to more complex values such as fairness, inclusivity, and ethical considerations. More powerful models have a greater potential to cause harm if not properly aligned, making alignment along the dimensions such as safety and ethical considerations increasingly important. In this section I will outline some of the most common methods for post-training alignment of large language models along with current limitations and other methods.

2.2.2 Alignment through supervised fine-tuning

A common first step of alignment is supervised learning where a dataset of prompts and expected response is used to fine-tune the pre-trained model. For most of preference optimization methods outline below supervised fine-tuning is used as the first step to create a base model [75, 82, 12, 60]. Depending on the dataset used the SFT step can help the model become better at following instructions [75], domain specific knowledge and style [63], induce Chain-of-Thought reasoning [105], change the values of the model [73], and many other capabilities. The key idea is that because the LLM already has a strong grasp of language and knowledge from pre-training, the SFT step can be used to nudge the model towards the desired behavior at a higher abstraction level than the predominantly low level token prediction task used in pre-training.

2.2.3 Alignment through preference optimization

Supervised fine tuning is useful however it has been shown that alone it doesn't sufficiently modify the behavior of the model. Preference optimization is a powerful next step which uses a dataset of prompts and potential responses that are ranked either by humans or AI models to indicate which responses are preferred over others. It is normally this preference data which is used to generate the supervised fine tuning dataset used in SFT by taking the highest ranked responses.

This preference data is usually (but not always) in the form of pairwise comparisons. That is given a prompt x , a pair of responses i and j the labellers (AI or human) indicate which

²ChatGPT which was the first widely used LLM was the fastest growing website ever[47].

³Other methods are proposed which embed the alignment within the pre-training step itself. Commonly used methods include filtering/curating training data.

response is preferred for example $i \succ j$ indicates response i is preferred over response j . If we have a dataset of these pairwise comparisons we can use the Bradley-Terry model [20] to create a maximum likelihood loss function

$$P(i \succ j | x) = \frac{\pi_i}{\pi_i + \pi_j}, \quad (2.1)$$

$$\mathcal{L}_{\text{BT}} = - \sum_{(x,i,j) \sim D} \log P(i \succ j | x) \quad (2.2)$$

$$= - \sum_{(x,i,j) \sim D} \log \frac{\pi_i}{\pi_i + \pi_j} \quad (2.3)$$

Where π_i and π_j are the scores of responses i and j respectively given prompt x . The way these scores are calculated depends on the specific preference optimization method used.

The success of preference optimization methods relies on the ability of modern LLMs to generalize learned preferences to new contexts and prompts not seen during training. The distribution of prompts in the preference dataset is crucial as it defines the context for which the model is expected to generalize its learned preferences. Therefore it is important to ensure that the prompts used to create the preference dataset are sufficiently representative of the contexts and dimensions in which you are expecting the model to align its behavior. Furthermore the rankings of the responses is also important as they define what the model is aligning towards. In many cases these rankings are subjective and provided by a small group of human labelers ($n \sim 50$). Therefore the alignment method may work well yet only reflects the values and preferences of a small group of people, which are usually the researchers and developers who give instructions to the labelers.

2.2.4 Reinforcement Learning from Human Feedback

Reinforcement Learning from Human Feedback (RLHF) [75] uses human labellers to create the preference dataset and then a reinforcement learning algorithm to optimize the model based on this feedback. This method popularized post-training alignment and was used to train InstructGPT which was smaller and better at following instructions compared to GPT-3 [21, 75].

How does RLHF work?

RLHF is separated into three distinct stages. Firstly is supervised fine-tuning (SFT) where a pre-trained language model is fine-tuned on a dataset of prompts and human written 'expected' responses to those prompts.

The second step is collecting multiple responses from the SFT model to create a set of prompts and response then having human labelers rank these responses from best to worst. This ranked data is then used to train a preference model (PM) that can predict which of two responses is better aligned to human preferences. The preference model is trained using a Bradley-Terry loss function where the scores π_i is equal to $\exp(\text{PM}_\theta(x, i))$. This is fed into 2.3 to create the loss function for the preference model.

Finally the fitted PM is used as a reward signal in the PPO [89] RL algorithm to further fine-tune the SFT model and create the final RLHF model.

Relevance

The idea of using human feedback to help train machine learning models is powerful, although it is not new and was around in theory and concept for years beforehand [26]. The

specific method of RLHF was widely adopted [98, 11, 93] yet is now replaced with more effective methods to utilize human feedback such as Direct Preference Optimization (DPO) [82].

2.2.5 Direct Preference Optimization

The idea presented in [75] introduces the concept of using human feedback to train large language models to be aligned to human preferences. However the process of RLHF is complex and requires training multiple models (SFT, PM, RLHF). Direct preference optimization (DPO) [82] is a method that simplifies this process by removing the need for reinforcement learning altogether. Instead DPO reformulates the preference modelling step to instead create an objective function that implicitly creates the reward function from the model and a reference model (usually the SFT model).

How does DPO work?

DPO starts with the same preferences data as RLHF; a set of prompts with responses ranked by humans. Like RLHF it starts with a pre-trained language model and is supervised fine-tuned on the prompts and highest ranked responses to create an SFT model. Instead of training a separate preference model DPO uses the SFT model as a reference model and defines the reward function $\hat{r}_\theta(x, y) = \log \pi_\theta(y|x) - \log \pi_{SFT}(y|x)$. Then by some derivation [82] it can be shown that the following loss function gradient can be used to optimize the Bradley Terry model directly on the preference data:

$$\nabla_\theta \mathcal{L}_{DPO}(\theta) = -\mathbb{E}_{(x, y^+, y^-) \sim D} \left[\underbrace{\sigma(\hat{r}_\theta(x, y^-) - \hat{r}_\theta(x, y^+))}_{\text{Reward model correctness}} \left(\underbrace{\nabla_\theta \log \pi_\theta(y^+|x)}_{\text{Increase preferred likelihood}} - \underbrace{\nabla_\theta \log \pi_\theta(y^-|x)}_{\text{Decrease dispreferred likelihood}} \right) \right]$$

Where σ is the sigmoid function, and (x, y^+, y^-) are the prompt, preferred response, and dispreferred response respectively. Rafailov et Al [82] found that the reward model correctness weighting term for the update is very important. This gradient update can be calculated with only 4 forward passes through the network per preference pair (2 for reference model and two for current model), making it much more efficient than RLHF ⁴.

Relevance

DPO provides a more effective way to utilize the same underlying feedback data that is used in RLHF. It replaced RLHF as the baseline method for post-training alignment. Due to its widespread adoption DPO represents a strong baseline for alignment using preference data.

2.2.6 Kahneman-Tversky Optimization

Traditional preference optimization methods such as RLHF and DPO set out with the objective of making the model maximize the likelihood of preferred responses over dispreferred responses. Ethayarajh et Al [36] sets out to change the objective of a language model to maximize the perceived utility of generations, which is how useful the generation is for the

⁴RLHF will need to do many forward passes of the network to train the preference model, then it will still have to do 3 forward passes for each datapoint when it comes to the RL step.

end user. The key being perceived utility is not the same as actual utility due to cognitive biases discussed in the follow sections. Ethayarajh et Al introduces the concept of Human-aware losses (HALOs) which optimize for what human perceive as the best outcome. It is shown that DPO and PPO are both HALOs which can help explain their success. Furthermore Ethayarajh et Al proposes a new objective function from the HALO family that works on only binary preference data⁵, making data collection easier and more natural.

Prospect Theory

KTO is built off the foundation of prospect theory [101] which is a theory to explain how humans make decision when dealing with uncertainty and risk⁶. It explains several cognitive biases that humans have when making decision involving risk, namely loss aversion (losses hit harder than gains), probability weighting (overweight small probabilities and underweight large probabilities), and reference dependence (we evaluate outcomes relative to a reference point which is usually the status quo). In relation to a LLM this means that when a human is evaluating the response of the model they are going to perceive the quality (utility) of the response in a biased way, so using a objective function that takes into account these biases (Human-aware) should lead to better alignment.

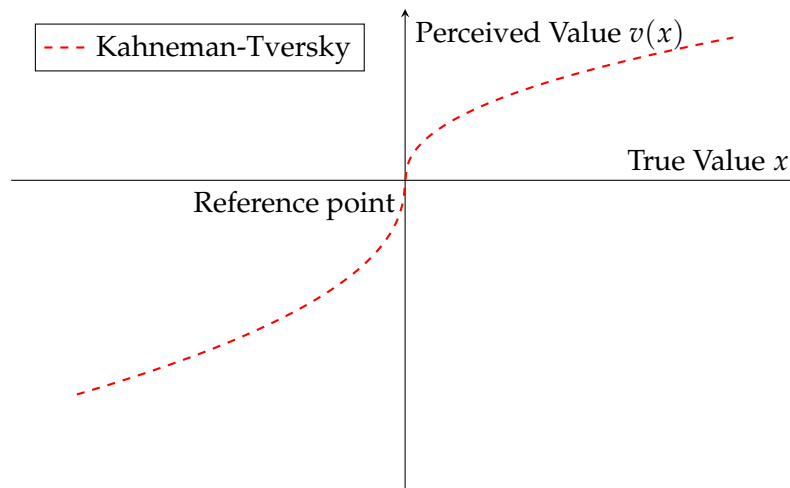


Figure 2.2: Implied human value functions showing loss aversion: concave for gains (risk averse) and convex for losses (risk seeking). Adapted from [101], with a change of terminology to match the context of value alignment and exaggeration for illustration.

Prospect theory does introduce several key concepts. Firstly is each random variable Z (the LLM output) has a utility which is determined by $\sum_{z \in Z} w(z)v(z - z_0)$. Secondly is the weighting function $w(z)$ which is used to weight the probabilities of each outcome z in a biased way. Finally is the value function $v(z - z_0)$ which determines how good/bad an outcome is compared to a reference point z_0 , this also will also include some bias from the individuals perspective.

⁵Given a prompt and response a labellers just need to label the response as desirable or undesirable

⁶For example why when humans are given the choice between \$50 and a 50% chance to win \$100, they often prefer the sure \$50, when they are both worth the same amount. Furthermore this effect is still present when the gamble is worth even more (\$110) [52]

How KTO works

KTO applies the idea of prospect theory by modifying the canonical utility function (shown in 2.2) provided in [101] to work for a language model fine tuning objective function. Namely it does this by omitting the weighting function (as users only see the final response, not the full probability distribution), making some minor adjustments for computational stability and using the Kullback-Leibler divergence [58] (between current model and reference model which is usually the SFT model) as a dynamic reference point. Notably this means that KTO only needs a database of prompts and responses labelled as either desirable or undesirable ($d = 1$ for desirable, $d = -1$ for undesirable) where the reference point is neutral (0). The KTO loss function is defined as:

$$\mathcal{L}_{\text{KTO}}(\pi_\theta, \pi_{\text{ref}}) = \mathbb{E}_{(x,y,d) \sim D} [\lambda(d) - v(x, y)]$$

where

$$v(x, y) = \begin{cases} \underbrace{\lambda_D \sigma(\beta(r_\theta(x, y) - z_0))}_{\text{increase probability above reference point}}, & \text{if } d = 1 \\ \underbrace{\lambda_U \sigma(\beta(z_0 - r_\theta(x, y)))}_{\text{decrease probability below reference point}}, & \text{if } d = -1 \end{cases}$$

$$\lambda(d) = \begin{cases} \lambda_D, & \text{if } d = 1 \\ \lambda_U, & \text{if } d = -1 \end{cases}$$

$$r_\theta(x, y) = \log \pi_\theta(y|x) - \log \pi_{\text{ref}}(y|x) \quad \text{Same reward model as in DPO}$$

$$z_0 = D_{\text{KL}}(\pi_\theta(y|x) \parallel \pi_{\text{ref}}(y|x)) \quad \text{Dynamic reference point}$$

The three hyperparameters β is a risk aversion parameter, and λ_D, λ_U are scaling parameters for desirable and undesirable responses respectively. In the references implementation and experiments $\lambda_D = \lambda_U = 1$ and the reference model is a SFT model.

Relevance

KTO offers several key logistical and theoretical advantages over prior preference optimization methods, that make it an attractive method for alignment. Firstly the logistical advantage. Collecting the ranked preference data which is needed by normal alignment methods is normally quite expensive [24] where binary like/dislike information is much more natural and less invasive to collect. KTO is shown to be more efficient and more robust to data imbalances than DPO. It is also less computationally intensive, a model of sufficient size (13B) does not need a SFT step prior to KTO so less training is needed and KTO has similar performance without using a reference model⁷ which halves the memory requirements when training. Secondly are the theoretical advantages. KTO is the first alignment method to explicitly take into account human cognitive biases when optimizing for alignment.

Another key point is the concept of applying prospect theory to alignment. Ethayarajh et Al apply it in the context of preference optimization, however prospect theory and the biases in human cognition can be applied more broadly than that in alignment.

⁷This is done by assuming that the reference model returns a uniform distribution across all outputs given x

There is something here yet I am not quite sure. One could change the axes in 2.2 to represent the concept of actual values and perceived values (however this feels like trying to shoehorn it in too much when the concept of revealed preferences is more appropriate). Another idea is that prospect theory explains how users will provide feedback in a biased way and we should actually be using the prospect theory to map that to the true thing we should align for, might help with generalisation.

2.2.7 Constitutional AI

What is Constitutional AI?

Constitutional AI (CAI) is a method of post-training a large language model to be aligned to a set of principles that are outlined in simple natural language document called a constitution [12]. Importantly it works without the need for human labelling of data common in other alignment methods such as RLHF and DPO [75, 82]. Instead of human labelling CAI uses the model itself to generate feedback on its outputs based on the principles outlined in the constitution, this feedback is then used to further train the model to align it to the constitution.

How does Constitutional AI work?

CAI is a process that takes in a constitution, a pre-trained language model, and a dataset of prompts and outputs a model that is aligned to the constitution. The process consists of three main steps: self generative and supervised fine-tuning (SFT), preference modelling (PM), and reinforcement learning with human feedback (RLHF) [12]. The constitution itself was generated in an adhoc manner by the authors of [12] and consists of a set of principles that aim to make the model's outputs more helpful, honest, and harmless.

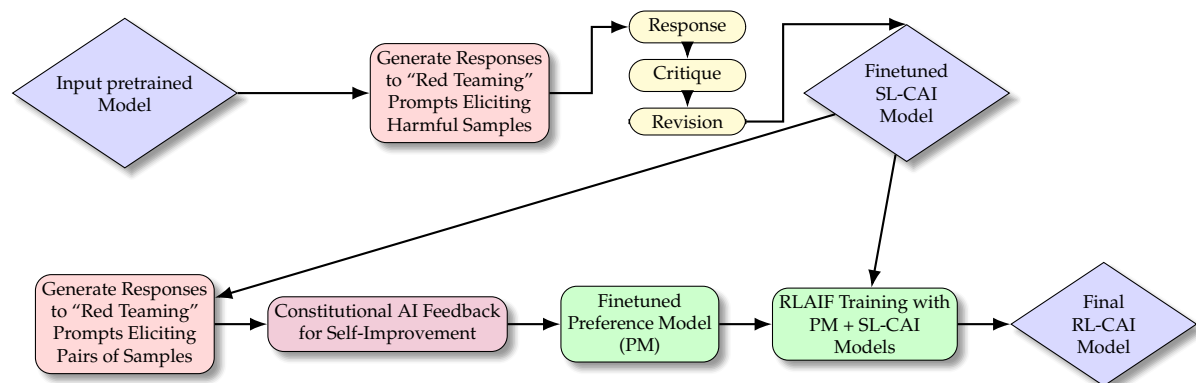


Figure 2.3: Overview of Constitutional AI process, adapted from [12]

Supervised learning on revised responses

The first stage of the process involves taking the pre-trained model and using it to generate responses to a set of prompts. These prompts are designed to elicit harmful behavior. As the pre-trained model is designed to be helpful it will likely generate harmful responses to these prompts. These responses are then critiqued by the model itself using principles outlined in the constitution, only a subset of which are used for each response critique. The critique is then used to revise the original response to make it more aligned to the constitution. This process of generating responses, critiquing them, and revising them is done iteratively to

create a dataset of revised responses. This dataset is then used to fine-tune the pre-trained model using self-supervised learning to create a model called the SL-CAI model.

Reinforcement learning with AI feedback

The next step involves using the SL-CAI model to generate pairs of responses to a set of prompts (overlapping with the set of prompts from the supervised step before or not). Then using the constitution the SL-CAI model provides feedback on which of the two responses is better aligned to the constitution. This feedback is then used to fine-tune a preference model (PM) that can predict which of two responses is better aligned to the constitution. Finally the PM and SL-CAI models are used together to do reinforcement learning in style of RFHL [75] to generate the final model RL-CAI. However the use of RLHF is not needed and other more modern methods like KTO and DPO can be employed. This final model is now aligned to the constitution without any human labelling of data. In the setup used in [12] the authors did in fact use some labels from human labelers which were used to provide the helpfulness signal in the preference model training, however in principle the preference signal could have been entirely generated by an AI model.

Relevance

In the context of when this paper was released there was a key dilemma of helpfulness vs harmlessness in large language models. Therefore, this paper set out to make a model that is both helpful and harmless, without sacrificing one for the other. Further research has expanded the horizons to be concerned with more than just harmfulness vs helpfulness trade off. As hinted at by the authors in [12], there is room for improvement in the CAI method to make it focus on more than just harmlessness vs helpfulness. This can be done by modifying the constitutions to include more diverse principles as well as modifying the red teaming prompts to target more diverse scenarios (i.e controversial topics, political leanings, etc). Collective Constitutional AI [48] is a direct extension of CAI and is the process of generating the constitution through public consultation.

2.2.8 Current technical alignment limitations

The goal of alignment is to ensure that you create a model that is aligned with what the designer and end users want. However there are several challenges that make this difficult namely being alignment data and effectiveness of alignment methods. Furthermore there is also a prevalent background concern of what/who the model is being aligned towards.

Preference optimization is the most used method of alignment which requires a dataset of prompts and preferred responses (ranked, or binary). There are several challenges with this data. Namely the cost of collecting it can be prohibitive [60, 24] and secondly is the concern of what/who the model is being aligned towards. For large scale post-training this dataset is normally collected from a small group of contracted labellers operating under the instructions of a small group of AI researchers and developers. As LLM model use becomes more prevalent it is becoming possible to organically collect preference data from real world users.

Current alignment methods demonstrate success on certain metrics that their methods work, that is they can make models more helpful, more honest, and less harmful [75, 12, 82, 36]. Yet the metrics used (broadly either human or AI evaluation of paired responses) miss out on deeper concerns around alignment. The first concern is that current methods may only achieve *shallow alignment* that is the model appears aligned during training yet fails to generalize this alignment to new contexts. Another concern is *inner alignment* where

the training process is aligned (outer-alignment) yet the model itself has learned objectives that are misaligned with the intended objective. Finally we observe models to make sure they perform how we expect them to, therefore we need *scalable oversight* methods to ensure that as models become more capable we can still ensure they are behaving correctly. These concerns are not new concerns in AI safety [6] yet they remain open problems in the context of large language models and their alignment.

The goal of alignment is to ensure that the model behaves and takes actions in a way that is aligned with what the designers and end users want. Current SOTA alignment methods are evaluated by how well the end model performs compared to the baseline judged by a AI or Human. The problem with this is that there is a significant gap between what we are measuring (short model outputs) and what we actually care about (long term model effects on the world). There are two arguments that support evaluating single prompt responses. One being pragmatic that it is much easier to evaluate short responses than long term effects. And secondly is on the theoretical grounds that any alignment goal no matter how large and complex can be reduced down into making the correct decision at each time step, where for a LLM a time step is a token generation or more broadly a response to a prompt. The theoretical argument relies on two assumptions that in practice don't hold up. Firstly is that we "the designers" can define what are the important action timesteps that need to be aligned, and secondly is that we can perfectly align the model at these crucial timesteps. Given that these two assumptions don't hold up in practice the long term effects of a model may not be aligned even if the short term outputs appear aligned. Therefore more work is needed to develop evaluation methods that can evaluate the alignment of the AI at the correct abstraction level and time scale.

2.3 Desiderata for AI

Alignment is a key concern in AI safety as we want to ensure that AI systems act and behave in a way that aligns with what humans want. More broadly we are at a transitional point in the history of Artificial Intelligence where the technology, systems and institution we build around AI have the potential to shape the future of humanity. Therefore it is paramount that we understand the context around AI, ways in which AI could bring about net negative change and the way in which we would like AI to be built and integrated into society.

2.3.1 Goal of AI

Anytime we add something to the world we should ensure that it is worth adding and going to provide at least net positive benefit to humanity. Measuring net positive benefit is hard and requires a metric that we can measure the positive and negative impacts of AI on humanity with. A complete discussion and outline of such a metric is out of scope for this section, furthermore a single perfect "goal" is not likely to exist [6] and if it did it wouldn't be a useful goal due to Goodhart's law [39]. However we can outline some high level ideas of what a good goal for the concept of AI could be, with the caveat and understanding that this is not a perfect goal and simply some high level guidance.

A good starting point for a "goal of AI" is the idea of human flourishing [103] that is split into 5 domains that are near universally agreed upon as good things for humans to have; happiness and life satisfaction, mental and physical health, meaning and purpose, character and virtue, close social relationships.

Definition 1 *The goal of AI should be to help maximize the median human flourishing across all*

people currently alive and in the future. ⁸

We can see the utility of this goal as it is broad enough to allow each individuals own interpretation of what human flourishing means to them, while still allowing us to understand harms and benefits of various AI systems. There is a valid concern that this measure of human flourishing is western culture centric and may not fully capture the values and needs of non-western cultures. More discussion is found in a later section on how to understand human values.

2.3.2 Stakeholders of AI

If we are on the precipice of a new technological revolution we need to understand what and importantly who is at stake. Artificial intelligence has the potential to impact everyone on the planet, an American firm that utilizes AI to handle customer service can take away jobs from urban workers in India. It can also have an impact through time, an AI paperclip factory that turns everyone into paperclips would extinguish humanity in the future and remove all possible positive futures for humanity. Ignoring certain groups of humans when designing and building AI systems leads to optimization for the benefit of one group over the other. Pressingly this happens when we ignore future people and take short term gains and push the costs and risks onto future generations or when we extract value from marginalized groups to benefit a select few ⁹.

There are many stakeholders of AI and to ensure that they are all represented we need to have the concepts of non-exclusion and non-domination. This means that no group of people should be excluded from the benefits of AI and no group of people's use of AI should dominate over another group of people's use of AI. When the groups are globally and temporally separated strict adherence to these principles could ensure that AI is at least evenly distributed across all people.

The problem however is when not all the stakeholders of AI are considered or empowered. Looking at the example of the customer service AI which will be designed, developed and deployed by a company for use by other companies. In a fair market it will make both companies rich at the expense of the workers who have lost their jobs. AI systems are overwhelmingly targeting companies as their 'users' which leads to companies driving the development and deployment of AI systems with disregard of the negative externalities on the true stakeholders which is humanity as a whole.

2.3.3 Alignment of AI

The actions of a few can impact the many. AI is a technology that can multiply the impact of one's actions. Therefore we need to ensure that everything around AI is aligned to the best interests of all stakeholders of AI. Yet due to no universal truth to govern what is best for all stakeholders we need to have AI that is aligned to the collective best interests of all stakeholders of AI. It is unclear what the method of collective alignment would be. Different subgroups of the stakeholders have different values and goals, aligning collectively therefore means finding a way to have a target which can be done in multiple ways.

1. **Pluralism:** Different groups will have different values. We have no ground truth to guide us on which values are better than the other. Therefore we should align models

⁸Including non humans at the root level of the goal is plausible. However one could reach the same conclusions of animal welfare by maximizing the human flourishing of "animal concerned" humans.

⁹Like modern day colonialism where the world's rich extract data, rare earth materials and labour from the world's poor and use it for their own gain. [30]

to be pluralistic in nature and make decisions that reflect this diversity along with the uncertainty about which values are most important. This can result in multiple “aligned” models or one model that can reflect different values in different contexts. Most aligned with the idea of agnostic democracy [106].

2. **Aggregation:** With the different sets of values from different groups we can aggregate these values to find a middle ground (naively by averaging) however this can lead to poor compromises and tyranny of the majority [70, 99]. Therefore more complex aggregation that weights different groups’ values by impact, vulnerability and importance can be used to find a better middle ground.
3. **Conglomeration:** If one assumes that all people can mostly agree then we can use ideas from deliberative democracy [41] to find a set of values that all people can agree on¹⁰. This is where methods like citizen assemblies and tools like polis [29] can be used to find a set of values that all people can agree on.

Regardless of the method of aligning to the stakeholders, the current goal of AI is misaligned with this definition as the goal of most AI systems is to maximize profit for companies and shareholders. It shouldn’t take excessive development to understand the negative externalities of profit motives [95]. The problem is not that GPT-4 itself is necessarily aligned to the goal of profit maximization (although future powerful AI systems could be), but that its second order effects (company shareholder profit) cause the development and deployment of the AI system to maximize profit. A related demonstration of this misalignment is the proliferations of recommender systems. These machine learning models that started with the intention of helping users find content they enjoy and want have turned into systems that prioritize engagement (and therefore ad revenue) over user flourishing [69]. We can see that the incentives of the institutions and frameworks that built the systems *captured* the original incentive of the AI system which could have been “help the user flourish”. This is why it is important not just to align the AI system itself, but also the institutions and systems around AI to be aligned to human values and goals. Full stack alignment [34] is this idea exactly where you need to make sure that individual models, government regulators and everything in the middle are all aligned to the target that we want.

2.3.4 How to do AI

If AI has any chance of achieving the goal of maximum average human flourishing we need to ensure that it is done in a way that is aligned to this goal. Full stack alignment is the idea of this however each part of the stack needs to be defined. A key idea here however is that due to objective capture the higher up the stack the alignment is the more influential the alignment is. Therefore aligning the institutions and frameworks which govern and build AI systems has the highest leverage for ensuring that AI is built in a way that is aligned to human values and goals.

Leaving AI development to private companies driven by profit might be effective in some ways and drive rapid development yet it is fundamentally misaligned with the goal of maximizing human flourishing (especially in the long term). A better framework around AI development is needed. Currently a democratic framework rules the world as the most progressive and forward thinking way to govern large groups of people.

“Democracy is the worst form of government, except for all the other forms that have been tried from time to time.”[27].

¹⁰All people agree can be operationally defined as “maximum equal approval” [50]

Even though democracy has its flaws it is arguably the most admissible way to govern. Therefore using democratic ideals and frameworks to also govern, spearhead and deploy AI is a good starting point for ensuring that the goal of AI is upheld. Operationally this means making artificial intelligence built by and for the public [81, 88, 76]. Various frameworks exist to help make AI both democratic and public.

Public AI provides a good that is both non-dominate and non-exclusionary. This method of making it a public good provides has three key features; public access, public accountability and permanent availability. Public access ensures that the opportunities to use AI and receive its benefits are available to all people. Public accountability crucially ensures that the stakeholders of AI (all people) have a say in how AI is developed and deployed, which allows AI the opportunity to attain the goal of maximizing median human flourishing. Finally the permanence feature allows it to be practically used and built up in much the same way that the electrical grid or internet is built and maintained.

There are two important failure modes however in this public approach. One is economic failure where the public good is not economically sustainable and therefore turns to undesirable methods of funding (ads, data extraction etc) ¹¹. The second is more of a global coordination/cooperation failure, even if all AI in the world is public AI one country could still prioritize its own interests over the collective good.

2.4 Problems of AI

Having an artificial intelligence that is aligned to human values and goals is paramount however it is not sufficient to ensure that AI is safe and beneficial for humanity. Furthermore there are several nuances and complexities in how AI will influence society that need to be understood and addressed to allow for a complete understanding on how to align AI effectively.

2.4.1 Concrete problems of AI safety

As much as in theory a perfectly aligned AI system would be as safe as possible, in practice there are several problems that arise when trying to align said system and what slight misalignment can cause. There have been multiple attempts in the last decade to outline and categorize these problems [6, 45]. I will outline some of the key problems here in a pseudo-categorization that can be understood as; we need AI to be competent and safe all the time, be trying to achieve what we want and be monitored effectively.

Firstly is **robustness** that ensures the model behaves reasonably at all times, specifically in situations that are out of distribution (adversarial, black swan, shifts etc) from the training data. A key challenge to this is that models mostly are unaware of when they are out of distribution therefore they must "always behave reasonably". This is easily implemented in a vacuum robot where if it is sufficiently confused it can just stop moving, however it is unclear how to implement this level of robustness in an AI doctor or lawyer as shutting down is not a reasonable response to a difficult situation. A promising direction to solve this problem is demonstrated phenomena that increasing model complexity increases its robustness [22]. At a high level this can be intuitive as a more complex model can contain and store more information about the world and therefore be able to generalize better to new situations. There has been limited evidence that this is true in modern LLM [46], with larger

¹¹Which is very much similar to the way that the internet was developed. It started off funded by the public sector (laying cables etc), however then for anyone to have a successful business offered on the internet it had to be "free" to use and so companies (E.g Google, Facebook) needed to find alternative methods of funding which has proven detrimental to greater public health

models having slightly more base robustness and being much more effective at receiving robustness training. Current models are still orders of magnitude away from the complexity of the human brain [86] and so may have much more “free” robustness to gain as they scale up in size and complexity.

Secondly is **alignment** which is ensuring that the model behaves in a way that is aligned to human values and goals. Given discussion above about this topic I will not further elaborate here. However it can be noted that alignment is plausible within a closed domain with well defined expected behavior (e.g preference datasets), yet it is still an open problem in open domains (e.g how should a politician act). A further key dimension to the openness of alignment is the temporal dimension. That is ensuring that the model remains aligned to our changing values and goals over time and prevent a situation of “value lock-in” where the model is aligned to a set of flawed values and beliefs ¹².

Lastly is **scalable oversight** which is ensuring we can keep track of and monitor the behavior of AI systems as they both scale up in capability and scale out in number of systems. If we had a provably aligned and robust system then oversight is not needed, however it is unlikely such a system will exist, instead we will likely have *demonstrably* aligned and robust systems. The only way to maintain confidence that these systems are behaving well is to continually monitor them. Doing this when they are highly capable presents several challenges due to how effectively they could deceive and/or how poorly humans could even understand what they are doing. Equally so when there are a large number of systems the available resources to monitor each agent decreases and so increasingly efficient and effective monitoring techniques will be needed. There are some promising results of bootstrap supervision [19, 35, 54] where AI systems are used to help monitor other AI systems. Although potentially quite fruitful this approach introduces new challenges of misalignment propagation as one needs to ensure that the supervising AI system is also robust and aligned.

2.4.2 Current negative impacts of AI

AI is already in the world and having an effect, there are many different types of AI harms that are having an effect in the near term. Common themes of negatives impacts of AI are as follows:

- **Bias and discrimination:** AI systems can perpetuate and amplify existing biases in society. For example in predictive policing [96], judicial sentencing [51] and hiring decision.
- **Automation and job loss:** AI systems can automate tasks that were previously done by humans, leading to job loss and economic disruption [9]. This can disproportionately affect certain groups of people, such as low-skilled workers and marginalized communities. Furthermore this leads to increased wealth concentration as the owners of companies that deploy AI systems reap the majority of the benefits.
- **Safety harms in high risk areas:** AI systems deployed in high risk areas such as healthcare [74], transportation and critical infrastructure can lead to real world safety harms if they fail or make incorrect decisions.
- **Population manipulation:** AI systems like recommender systems, online advertising and social media algorithms can manipulate public opinion and behaviour [zuboffAgeSurveillanceCapitalism2019, 4]. This has had some terrible real world consequences such as the Rohingya genocide in Myanmar [83]

¹²For example training a model on 1820 European values might lead a model to reinforce the racist and sexist beliefs of that time period.

- **Explicit misuse:** AI systems that are more capable are more at risk of being used for malicious purposes which is called the dual-use problem. Most common in dual-use concerns are Bio weapon design and cyber warfare [97, 87].

Many of these harms come about due to either negligent or malicious deployment of AI systems. Where having systems that are better aligned to human values could help mitigate some of these harms. However there are also systemic issues with the way that AI is developed and deployed that lead to these harms. Namely the profit motive of private companies is not aligned with the goal of "helping users being their best self". Therefore solutions to these problems are the same solutions to how we stop companies in general from causing harm, namely regulation, oversight and changing the incentives of companies.

2.4.3 Catastrophic and existential risks of AI

The stakes are high with AI. There are multiple plausible scenarios where AI could cause human extinction or catastrophic risks to humanity. We may be living in a very vulnerable world where the majority of possible future states are negative from our current perspectives [18, 66]. Central to these paths and scenarios is the idea that more powerful and capable AI systems have the potential to enfeeble humans and disempower us. Once in this state of disempowerment humans are by definition unable to control and direct their own future. Therefore if any form of misalignment exists between the AI system and the maximum human flourishing goal the consequences could be catastrophic or existential.

AI development and AI progress has increased rapidly in the last few years [59]. It has increased in speed due to scaling laws of AI [53] and increased investment from both private and public sector. The key next step is that AI systems will accelerate and automate many parts of AI research. This could result in an intelligence explosion [17, 94] where the input that humans play in AI development tends towards zero and improvement can happen on timescales much faster than humans. Even if this intelligence explosion happens over the course of many years it is still much faster than human development over the last few thousand years. Therefore the probability of highly capable AI systems could be much higher than previously thought assuming AI can appreciably automate its own development.

With these highly capable AI systems there are several plausible catastrophic risks that could occur. These can generally be split up into two sorts of catastrophic risk, **rogue AI** and **malicious use of AI** [44]. Rogue AI is where a system is either deceptively, overtly or indifferently doing actions against the goal of maximizing median human flourishing. The impact of this risk increases as the capabilities and proliferation of AI systems increases. The second risk also has similar increasing impact, where bad actors can do increasingly harmful things with increasingly powerful and accessible systems (bioweapons, cyber warfare etc). Either of these situations have the potential to cause an irreversible catastrophic risk to humanity. This could be through direct extinction of humanity, irreversible collapse of civilization or permanent disempowerment of humanity [17].

As the impact of these risks increases the probability of them can also be increased through both organizational incompetence (e.g. The USSR and Chernobyl, NASA and the Challenger disaster) and competitive pressures (e.g. Nuclear arms race) [44]. These two factors broadly lead to decreased safety standards with increased focus on rapid development and deployment.

2.4.4 Preventing catastrophic and existential risks of AI

Given the stakes are very high there should be significant effort into ensuring that these risks are mitigated and negative events avoided. One can think about AI safety and ways of

preventing catastrophic risks as gas pedal (constructive) and brake (restrictive) actions.

The first common approach is a halt or moratorium on AI development. Although reasonable sounding at first glance relatively large scale attempts at this have failed [77]. A global pause struggles from coordination problem where each actor has no incentive to pause as others might not pause and therefore gain competitive advantage. Furthermore pausing in an indefinite manner runs into the ethical problem about forgoing potential benefits of AI ¹³. This is why most pauses call for a limited time period to allow for safety, policy and governance research and implementations to catch up [78]. Given the nature of AI development requiring large amounts of resources (hardware, data, power, etc) it is plausible that a well designed pause and global coordination could be effective. It could work much like nuclear non-proliferation treaties where countries agree to limit their development of nuclear weapons and allow for inspections to ensure compliance. Also much like nuclear weapons most of current day frontier AI development requires massive data centres that are hard to hide and therefore easier to monitor, much like uranium refinement factories.

If one had time what would be the methods and ways to reduce the risks? Generally speaking they fall into technical safety solutions and governance and policy solutions.

There are proposals for having various policies and regulations that would slow down the progress of AI development. Namely so that we have more time to understand the systems and develop appropriate safety measures. Policies for slowdown can involve computation restrictions, increased liability for the developers and sellers of AI systems as well as more stringent testing and evaluation requirements. Some of these can help long term safety however most of them are simply mandating a slower process and better appreciation of the risks.

With this better appreciation of the risks we can implement further policies and also develop novel technical solutions. There are various methods for AI that are safe by design [32, 16]. The key to these methods is that the resultant AI is either deeply aligned by construction or is limited in capability to prevent misalignment from causing catastrophic risks.

2.5 Representing Human Values for LLM alignment

As discussed in the previous sections it is paramount that we make sure that large language models (LLMs) are aligned to human values and by extension behavior. There remains a key questions of how do we represent and understand human values that would allow us to align a AI system to them. There are broadly two methods one works from defining broad principles and specific actions are inferred from there the other method is providing specific examples and generalizing from them. Interestingly when working with humans we can only provide principles and let them infer the details, yet with modern AI systems like LLMs we can provide lots of specific examples and let it infer the values from this ¹⁴. In this section I will outline some of the key concepts around representing human values and how they relate to AI systems.

¹³This concept of whether we do need AI and the importance of bringing it to fruition compared to its potential downsides is a very important question that shouldn't be glossed over too much. However in the context of AI alignment it is outside the scope and thus not discussed further.

¹⁴Providing a human with 10,000 different questions and responses of "how to answer the phone" won't provide much clarity, yet giving them some simple high level rules is quite effective. Oppositely for AI most alignment methods like RLHF, DPO etc provide lots of specific examples and let the model infer the underlying values.

2.5.1 Human values taxonomy

Human values are an old idea that go back millennia [7, 71, 5], with each culture having its own set of values that are important to them. They define how people should be and act ¹⁵. In modern times we have developed various taxonomies of human values that allow us to understand and categorize different sets of values.

A widely cited modern value taxonomy is Rokeach's book "The Nature of Human Values" [84] where he posits that human values can be understood as 36 distinct values. Rokeach groups the values as either *terminal values* which are desirable end-states (freedom, equality...) and *instrumental values* which are preferable modes of behavior (honesty, kindness...). Expanding on this idea of value taxonomy Schwartz [90] proposes a model of 10 broad human values that he later expands to 19 values [92]. Schwartz organises the values in a circumplex structure where one dimension is self-enhancement vs self-transcendence and the other is openness to change vs conservation.

These taxonomies provide a useful framework for understanding human values and how they relate to each other. Importantly they are trying to provide the most predictive ability of people's actions given the least number of values. This trade off between complexity and predictive power is important when trying to represent human values in AI systems. Contemporary work from computer scientists have created value taxonomies that are designed specifically for their predictive ¹⁶ power [55]. Yet we can see that it is more complex (54 values) and not as interpretable as the psychological taxonomies. This means that it works well for AI systems yet is hard to apply to humans. This means that for the purpose of AI alignment we need a taxonomy that is both interpretable and has good predictive power that would allow us to align AI systems to human values effectively.

2.5.2 Value surveys

One of the key features of early modern value taxonomies is that they were created with associated survey instruments that allowed researchers to measure the values of individuals. Rokeach [84] created the Rokeach Value Survey (RVS) which asked participants to rank the 36 values in order of importance to them. Schwartz [90] created the Schwartz Value Survey (SVS) which asked participants to rate the importance of each value on a 9-point scale. Later Schwartz replaced the SVS with the Portrait Values Questionnaire (PVQ) [91] which was less abstract and better at soliciting the values from a diverse set of people as respondents are answering questions of the form "How much is this person like you", and by rating their similarity with 40 different portraits one can infer someone's values. These surveys allowed researchers to measure the values of individuals and groups and provided a way to understand how values relate to behavior.

All the above surveys (PVQ, SVS and RVS) are designed to measure the values of individuals in relation to a predefined taxonomy of values. They have relatively few questions (< 50), are abstract in nature and quick to complete. An alternative path for measuring values is to create broader surveys that try to capture the values of large groups of people through many questions. Importantly these surveys do not map directly to a predefined taxonomy of values yet instead try to capture the complexity of human values through many questions (300). The World Values Survey [43] is the biggest example of this where they have been surveying over 1,000 people from over 100 countries every 5 years since 1981. The New Zealand Attitudes and Values Study (NZAVS) [102] is another example of a large scale value

¹⁵Although the actual definition of values is up for contention, this simple version can suffice for our purposes

¹⁶Which in this case is the ability for a model to predict someone's values (from the taxonomy) given an argument that they made. Plausibly the word modelling power might be more intuitive.

Provide a reason-
ing as to why
inter-
pretabil-
ity is im-
portant
here.

survey that has been running since 2009 surveying around 30,000 New Zealanders every year. These surveys don't correspond to a strict values taxonomy and instead provide granular data from each individual. This data can help train [73, 62] and evaluate [33, 107, 8] Large Language Models (LLMs) on human values.

These surveys provide granular data yet struggle from well documented issues of self-reporting bias [80] and social desirability bias [31]. This means that the values reported by respondents may not accurately reflect their true values. From an economics perspective this is the distinction between stated values (ones that are reported in surveys) and revealed values (ones that are implied by actions an agent takes). This provides a caveat and limitation to using value surveys as a way to represent human values for AI alignment.

2.5.3 Values to behavior

For the purposes of building AI systems understanding values is only useful as far as they can predict behavior (in the form of responses and actions). That is we want the AI system to behave the same or better in all situations that a human would. There is contention in psychology and AI alignment research around the values to behavior gap [57, 37]. When one tries to bridge the gap between understanding someones behavior and values we run into common problems namely; non-identifiability of values (Many different values can lead to the same behavior), specification gaming (as mentioned above) and extrapolation errors. Non-identifiability and extrapolation errors are demonstrated clearly in inverse reinforcement learning (IRL) [1]. Furthermore it can be shown that human values are fickle and can stem from ones previous actions as humans rationalize what they have done [61]. This is the reasons as to why many AI alignment researchers focus on direct behavior alignment through revealed values with pairwise preference ranking methods as discussed above like DPO, RLHF and KTO.

Use some of the reference from [56] and other research to talk more about values and different definitions.

However the idea that values can lead to and predict behavior has been argued for [85, 91]. Furthermore in the reinforcement learning literature there is evidence that given examples of actions a model can learn a opaque value function that can predict what actions will be made next (called inverse reinforcement learning). The methods that are most effective for achieving this is through combining multiple signals (stated values, observed behavior and explicit feedback) [2]. In machine learning this is done through collaborative learning where different types of value signals at different times are used to train/fine-tune a model [42, 75]. Using multiple signal also help mitigate the gap between true values and stated values. Ideally one would just align a model to a persons values then behavior would follow. However due to the gap between values and behavior one needs to provide additional feedback to the model which for LLM is done through preference optimization methods outlined above.

2.5.4 Why human values are important for AI alignment

The is-ought problem is an old philosophical concept that states that one cannot decide what ought to be purely from what currently is [49]. This presents a challenge for AI alignment, as we must determine what values we ought to align AI systems to, and all we currently have is what human values currently are. To align towards current human values is making a prescriptive statement on what ought to be human values given purely off what we currently have, which as per Hume's Law is inadmissible. Yet given the infeasible nature of other value targets using democratic ideals is the most defensible approach. That is to say the

distribution of human values is the best target we have to align AI systems to, and by avoiding value lock-in we can allow us to adapt and grow. Alternative methods like inaction—letting the model be unaligned and just be what it is— or aligning to a specific set of values (e.g. expert values) are fraught with issues of bias, lack of representation, and potential for harm. Therefore, aligning AI systems to human values (specifically democratically collected values) is the most defensible and thus ethical approach to AI alignment.

Chapter 3

Project Proposal

3.1 What I will do

The goal of this project is to develop, implement and evaluate for future feasibility a method for aligning large language models to a set of representatively collected human values. This will be an end-to-end project that involves using current survey data and collecting new data, aligning a large language model using different methods and evaluating the effectiveness of the alignment.

3.2 What problem I will solve

Current methods for training large language models involve scraping the internet for large amount of text data. This creates a very powerful next token prediction model. Then to turn this models into something useful for the end user (e.g. chatbots, assistants, robo doctor etc) the model is aligned to how we want it to behave. Current methods for alignment involve contracting human labelers/annotators/demonstrators to provide data to give to the model on how we want it to behave. Problematically this is a opaque process that gives the end user little insight into what the model is "designed to do". Instead we can let the public directly and indirectly decide on what values they want the model to be aligned to. More so the public as the users and recipients of the model's outputs should have a say in the way that the model behaves. Large language models are on track to become tomorrows, accountants, lawyers, doctors and more, we need to ensure that we the people are the ones who decide on what kind of people they are.

3.3 Timeline

For now I will leave the timeline uncompleted. Once I have a more concrete idea of the method I will fill this in properly. There is code left in to help with how to make a gantt chart.

The project will be carried out over three main phases. The first phase will be the organization and clustering of existing survey data to represent human values. The second phase will be an iterative process of aligning a large language model and evaluating its performance. The third phase involves collecting specific alignment data to further refine and train the model, followed by a final evaluation.

3.4 Output

The output is both qualitative and quantitative in nature. There are four main deliverables:

- A collection of aligned LLMs with various checkpoints of just survey data and then with the collected data.
- A public dataset of collected data around preferred values and responses for alignment.
- Experimental results evaluating the effectiveness of different alignment methods using the collected data.

3.5 Resources needed

There are two resource requirements for this project, hardware and public survey responses.

3.5.1 Hardware

The first is hardware which will be needed to do the alignment finetuning evaluation of the models. For the base LLM I would want to use a model in the range of 7 billion to 30 billion parameters, as this will get me close to state of the art performance on many tasks. This means that I will need access to about 200GB of GPU memory to be able to finetune the model effectively. Each finetuning run will take in the realm of 3-6 hours depending on the size of the model, amount of data used, alignment methods used and GPU hardware. Therefore given the goal of trying a handful of methods and different value sets I estimate about ¹ 70 hours of GPU time will be needed. The cost of a GPU hour on a high end GPU such as a B200 is 6 USD per hour which would make the total cost about 420 USD. Alternatively the Engineering and Computer Science department's GPU servers should give me access to machines with around 80GB of vram. This would allow me to fine-tune smaller models in the 7-13 billion parameter range. This should still be sufficient for a proof of concept and evaluation of the alignment methods.

Public consultation responses

The idea of using the public to help align large language models is central to this project. Most of the fine tuning will be done using the survey data which can be used to get the model roughly aligned to human values. However to get more specific and granular alignment I will collect new data from the public to further refine the model's behavior. This will be in the form of an interactive survey where participants will provide some form of preference feedback on model outputs. I envision using a tool like Polis or similar to collect and aggregate responses in a meaningful way. I would expect to need at least 50 or so people from the various subgroups and for each of them to spend maybe 2-3 15 minutes blocks providing feedback on model outputs.

¹4 hours per fine tune * 3 different methods * 5 different value sets + 5 hours of evaluation = 65 hours

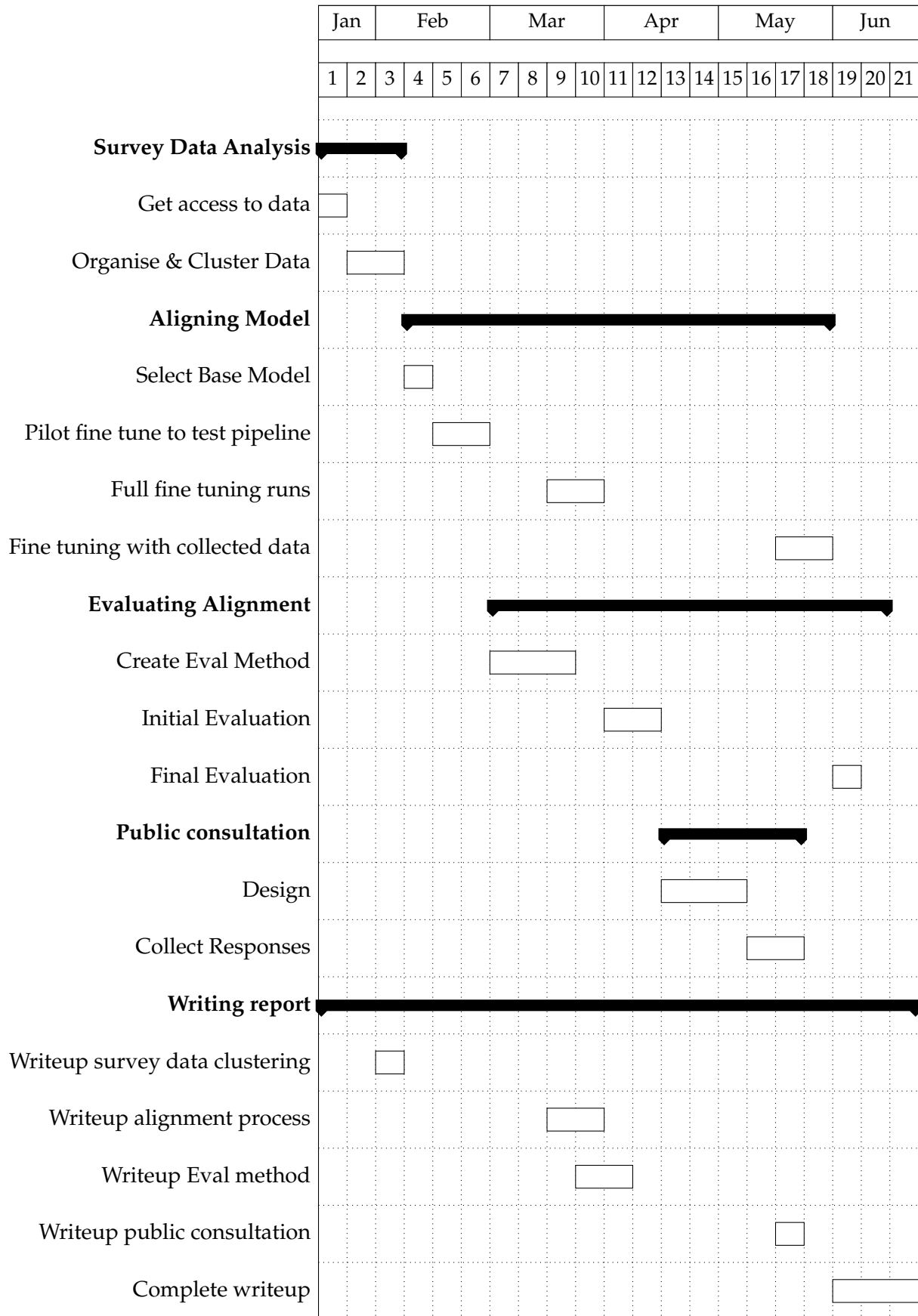


Figure 3.1: Project timeline in weeks (only roughly aligned to months). This assumes a start time of end of January 2026 with near part time work ramping up to full time work by end of February.

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